Engineering Mathematics: Mastering the Chain Rule

Question 1

Problem Statement

Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ if:

- $w = x + 2y + z^2$
- $x = \frac{r}{s}$
- $y = r^2 + \log(s)$
- $z = 2r$

Dependency Analysis

- The primary function w depends directly on three intermediate variables: x, y, and z.
- Each of these intermediate variables depends on the independent variables r and s.
- Therefore, to find a derivative with respect to r or s, we must sum the contributions passing through all three intermediate channels (x, y, and z).

```
        w
      / | \
     /  |  \
    x   y   z
   / \ / \ / \
  r   s r  s r  s
```

Let's Establish the Building Blocks

Before building the full paths, let's find the individual derivative links.

Top-to-Intermediate Layer (Differentiating w):

- $\frac{\partial w}{\partial x} = 1$
- $\frac{\partial w}{\partial y} = 2$
- $\frac{\partial w}{\partial z} = 2z$

Intermediate-to-Bottom Layer (Differentiating with respect to r and s):

For $x = r \cdot s^{-1}$:

- $\frac{\partial x}{\partial r} = \frac{1}{s}$
- $\frac{\partial x}{\partial s} = -\frac{r}{s^2}$

For $y = r^2 + \log(s)$:

- $\frac{\partial y}{\partial r} = 2r$
- $\frac{\partial y}{\partial s} = \frac{1}{s}$

For $z = 2r$:

- $\frac{\partial z}{\partial r} = 2$

- $\frac{\partial z}{\partial s} = 0$ *(Since z has no s terms, it behaves as a pure constant)*

Step-by-Step Execution

1. Finding $\frac{\partial w}{\partial r}$

Sum the total derivative contributions through x, y, and z with respect to r:

$$\frac{\partial w}{\partial r} = \left(\frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r}\right) + \left(\frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r}\right) + \left(\frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial r}\right)$$

Substitute the building blocks into the expansion formula:

$$\frac{\partial w}{\partial r} = (1) \cdot \left(\frac{1}{s}\right) + (2) \cdot (2r) + (2z) \cdot (2)$$

$$\frac{\partial w}{\partial r} = \frac{1}{s} + 4r + 4z$$

Since the problem requires the answer **in terms of r and s**, substitute $z = 2r$ back into the equation:

$$\frac{\partial w}{\partial r} = \frac{1}{s} + 4r + 4(2r)$$

$$\frac{\partial w}{\partial r} = \frac{1}{s} + 12r$$

2. Finding $\frac{\partial w}{\partial s}$

Sum the total derivative contributions through x, y, and z with respect to s:

$$\frac{\partial w}{\partial s} = \left(\frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s}\right) + \left(\frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}\right) + \left(\frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s}\right)$$

$$\frac{\partial w}{\partial s} = (1) \cdot \left(-\frac{r}{s^2}\right) + (2) \cdot \left(\frac{1}{s}\right) + (2z) \cdot (0)$$

$$\frac{\partial w}{\partial s} = -\frac{r}{s^2} + \frac{2}{s}$$

Question 2

Problem Statement

If $z = f(x, y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, prove that:

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$$

Dependency Analysis

- The dependent variable z is a function of intermediate variables x and y.
- Both x and y are functions of independent variables u and v.
- Any derivative with respect to u or v must split into two parallel tracks: one through x and one through y.

```
      z
     / \
    x   y
   / \ / \
  u   v u   v
```

Let's Establish the Building Blocks

Since $z = f(x, y)$ is an unspecified function, we leave $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ intact as operational terms. Let's compute the underlying variable relationships:

- $\frac{\partial x}{\partial u} = \frac{\partial}{\partial u}(e^u + e^{-v}) = e^u$
- $\frac{\partial x}{\partial v} = \frac{\partial}{\partial v}(e^u + e^{-v}) = -e^{-v}$
- $\frac{\partial y}{\partial u} = \frac{\partial}{\partial u}(e^{-u} - e^v) = -e^{-u}$
- $\frac{\partial y}{\partial v} = \frac{\partial}{\partial v}(e^{-u} - e^v) = -e^v$

Step-by-Step Proof

Step 1: Write out the total derivative expansions for the LHS

Using the chain rule expansions, write out the expressions for $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = \frac{\partial z}{\partial x}(e^u) + \frac{\partial z}{\partial y}(-e^{-u})$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} = \frac{\partial z}{\partial x}(-e^{-v}) + \frac{\partial z}{\partial y}(-e^v)$$

Step 2: Evaluate Left-Hand Side ($\text{LHS} = \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v}$)

Subtract the second expansion expression from the first expression:

$$\text{LHS} = \left[e^u \frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y}\right] - \left[-e^{-v}\frac{\partial z}{\partial x} - e^v \frac{\partial z}{\partial y}\right]$$

Distribute the negative sign carefully across both terms:

$$\text{LHS} = e^u \frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y} + e^{-v}\frac{\partial z}{\partial x} + e^v \frac{\partial z}{\partial y}$$

Group the common derivative factors together:

$$\text{LHS} = (e^u + e^{-v})\frac{\partial z}{\partial x} + (e^v - e^{-u})\frac{\partial z}{\partial y}$$

Step 3: Compare with the Right-Hand Side (RHS)

Refer back to your original structural variables given in the prompt:

- $x = e^u + e^{-v}$
- $-y = -(e^{-u} - e^v) = e^v - e^{-u}$

Substitute these exact variable values directly back into your grouped expression:

$$\text{LHS} = (x)\frac{\partial z}{\partial x} + (-y)\frac{\partial z}{\partial y}$$

$$\mathrm{LHS} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = \mathrm{RHS}$$

This completes the proof.